End Semester Examination 2025-26

Name of the Program: **M. Tech (CSE)** *Semester:* ***I***

Name of the Course: **Advance Operating Systems** *Course Code:* **MCS-103**

Time: **3 Hours** Maximum Marks: **100**

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	**(20 marks)**	
a)	**Discuss some of the reasons for implementing process migration in distributed environment.**	CO1
b)	**What are different types of schedulers. Explain working of each scheduler. Why it is important for a scheduler to distinguish between I/O-bound programs from CPU-bound programs?**	
c)	**Describe operating system with the help of its basic structure. Explain various functions of operating system.**	
Q2	**(20 marks)**	
a)	**Most round-robin schedulers use a fixed size quantum.** **a. Give an argument in favor of a small quantum.** **b. Now give an argument in favor of a large quantum.** **c. Compare and contrast the types of systems and jobs to which the arguments apply. Are there any for which both are reasonable?**	CO2
b)	**Explain distributed snapshot algorithm by taking suitable example.**	
c)	**Explain semaphore. Compare and contrast the performance of busy-waiting and without busy waiting semaphores in different scenarios.**	
Q3	**(20 marks)**	
a)	**Discuss situations under which the least frequently used page-replacement algorithm generates fewer page faults than the least recently used page replacement algorithm. Also discuss under what circumstance does the opposite holds.**	CO3
b)	**How dead lock is detected in Distributed environment. Explain various Distributed Deadlock Detection Strategies**	
c)	**Consider a preemptive priority scheduling algorithm based on dynamically changing priorities. Larger priority numbers imply higher priority. When a process is waiting for the CPU (in the ready queue, but not running), its priority changes at a rate A; when it is running, its priority changes at a rate B. All processes are given a priority of 0 when they enter the ready queue. The parameters A and B can be set to give many different scheduling algorithms. What is the algorithm that results from B > A > 0?**	
Q4	**(20 marks)**	
a)	**Explain and critically compare the Monolithic kernel architecture and the Microkernel architecture. Your answer should detail the fundamental structure of each, discuss at least two distinct advantages and two distinct disadvantages for each architecture, and conclude with a discussion on which architecture is generally preferred for modern, resource-constrained, embedded systems and why.**	CO4
b)	**Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order, is**	

	86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130 Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for SSTF and C-LOOK disk-scheduling algorithms?	
c)	Consider a single level paging scheme. The virtual address space is 256 MB and page table entry size is 4 bytes. What is the minimum page size possible such that the entire page table fits well in one page?	
Q5	(20 marks)	

a)	Suppose your friend proposed solution against the three requirements for a valid Critical Section solution. He has implemented following solution to the problem for two processes P0 and P1, where flag[2] is a Boolean array of two elements and turn is an integer variable. Both are global to processes P0 and P1.
	Is this solution is correct? Justify your answer.
	bool flag[2]; // global flags
	int turn; // global turn variable
	Process i (i = 0 or 1)
	flag[i] = TRUE; // I am interested
	while (turn != i) { // while it is not my turn
	while (flag[J]) // while you are interested
	; // do nothing: busy waiting
	turn = i; //because you are not interested, it is my turn
	}
	// critical section//
	flag[i] = FALSE // I am done and not interested	

b)	A computer system uses a three-level paging scheme to translate logical addresses to physical addresses. The system is configured such that a main memory access takes 200 ns. It is observed that the Effective Memory Access Time (EAT) is 650 ns for address translation and data retrieval. Assume the following conditions: 1. There is no TLB (Translation Lookaside Buffer) in use. 2. There are no page faults (all pages are in main memory). 3. Each level of the page table (the Page Directory, the intermediate page table, and the final page table) requires one separate main memory access to fetch the next level's base address or the final page frame number. What is the memory access time (in ns) required to fetch the final data word from the physical address, once the translation is complete? (Assume the time taken for all CPU operations, like addition and comparison, is negligible.)	CO5
c)	Show a scenario (using an example of resource access or process scheduling) where deadlock detection and recovery mechanisms (e.g., process termination) could unintentionally lead to or exacerbate starvation for a repeatedly chosen victim process. Conversely, explain how a mechanism designed to prevent starvation (like aging) could potentially increase the risk of deadlock if not carefully implemented alongside a deadlock avoidance or detection scheme.	